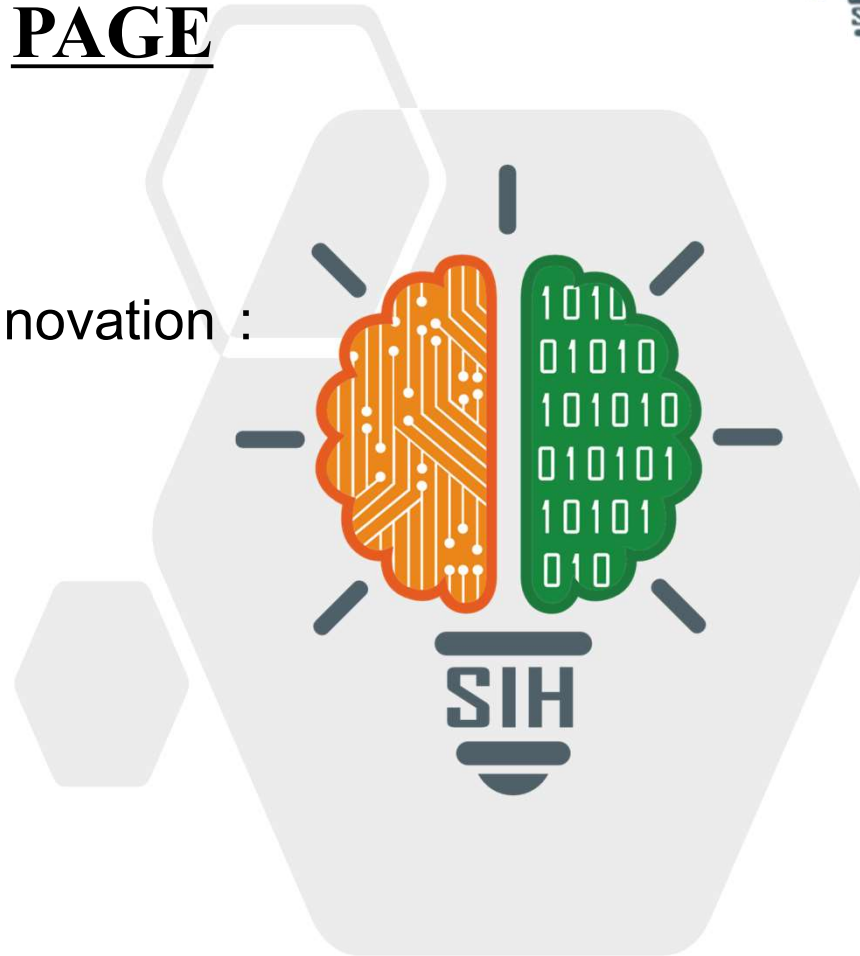


# SMART INDIA HACKATHON 2025



## TITLE PAGE

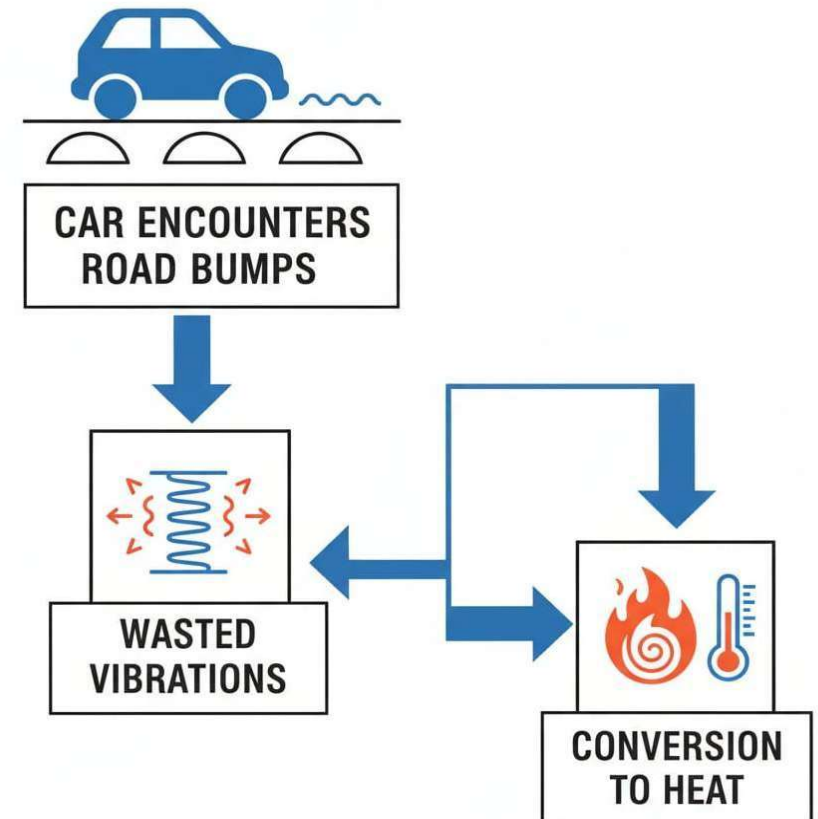
- **Problem Statement ID**-SIH25119
- **Problem Statement Title**-Student Innovation :  
Atmanirbhar Bharat Smart Vehicles
- **Theme**-Smart Vehicles
- **APS Category**- Hardware
- **Team ID**-
- **Team Name**-AutoTron



# Energy Harvesting Suspension System using Linear Generator

## ❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- Converts suspension motion → electricity
- **Dual purpose:** shock absorption + energy harvesting
- Stores power for vehicle electronics/EV charging
- Eco-friendly, cost-effective, innovative

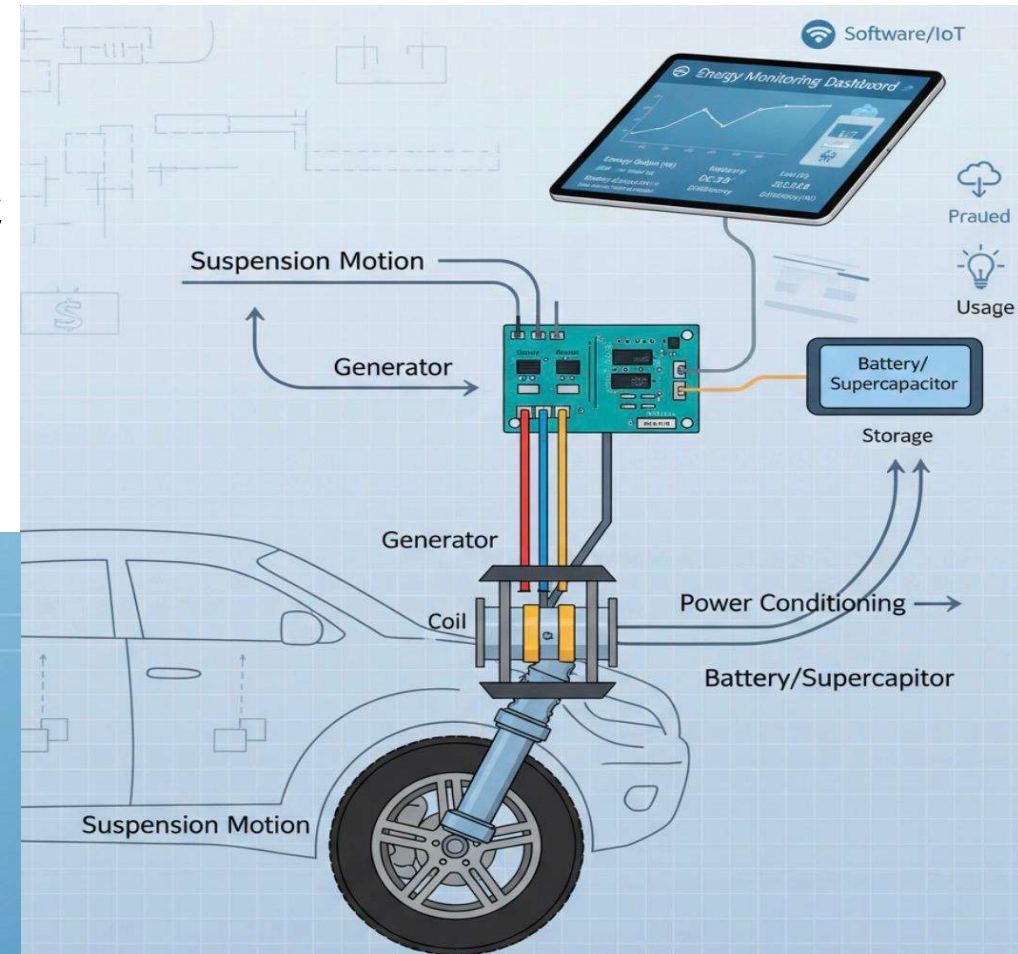
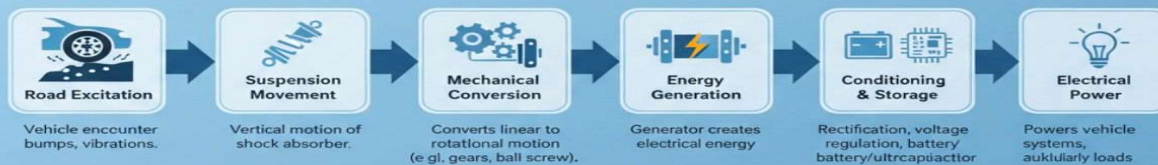


# TECHNICAL APPROACH

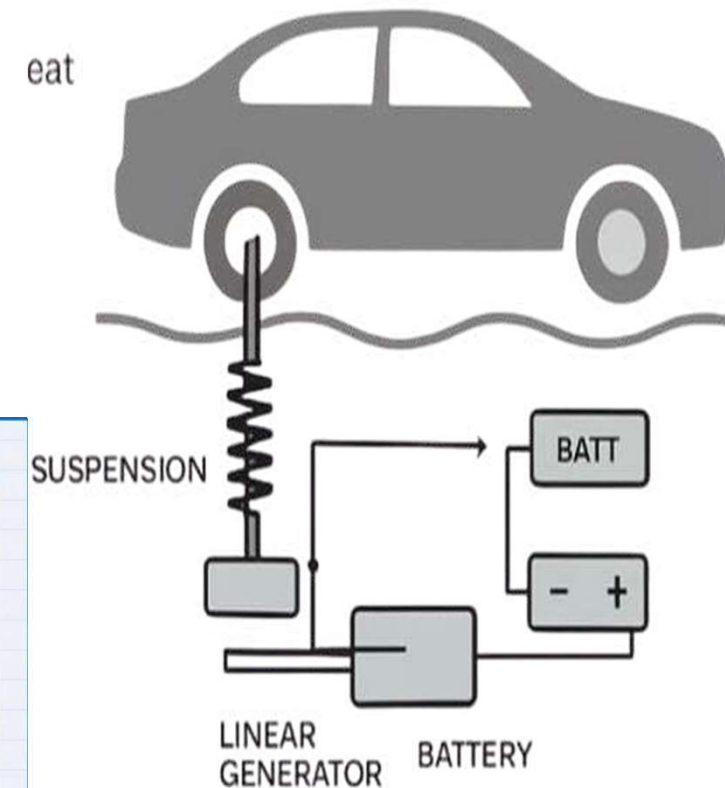
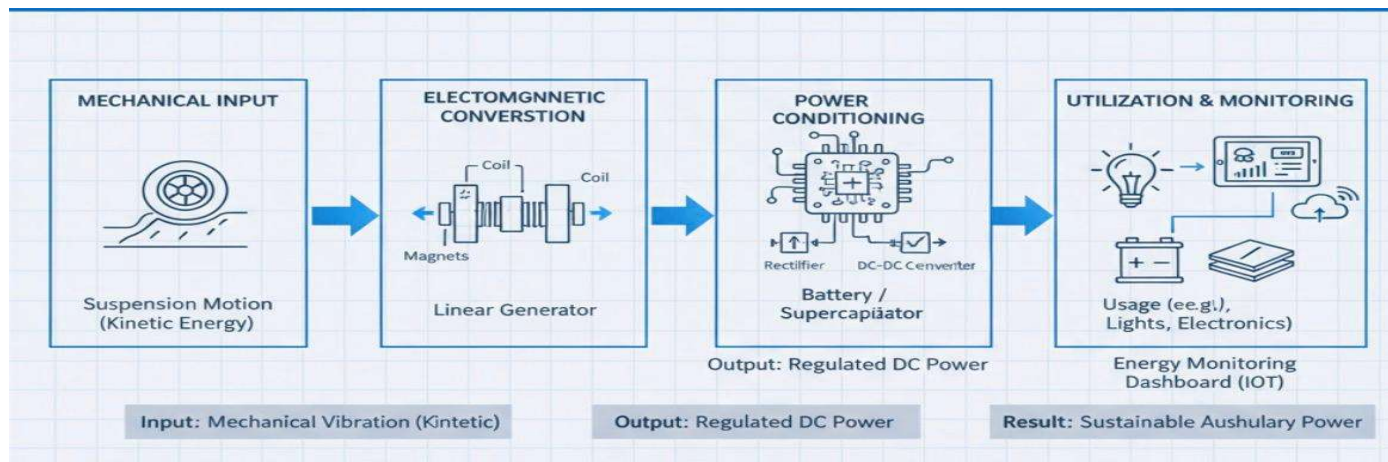


- **Hardware:** Linear generator, coil, magnets, suspension unit
- **Electronics:** Rectifier, DC-DC converter, battery/supercapacitor
- **Software/IoT:** Energy monitoring dashboard

## Energy Harvesting Suspension: Process Flow



- **Feasible:** Simple electromagnetic principle
- **Prototype:** Ready in hackathon
- **Challenges:** Efficiency, weight, durability
- **Solutions:** Optimized design, lightweight alloys, protective casing



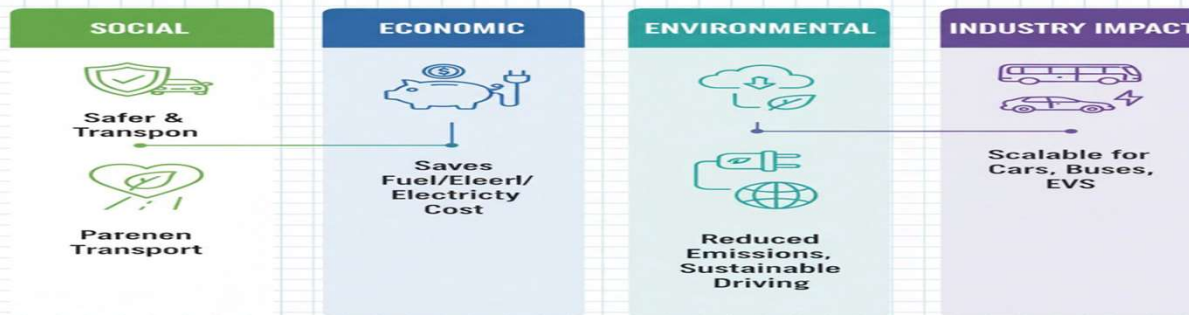


# IMPACT AND BENEFITS



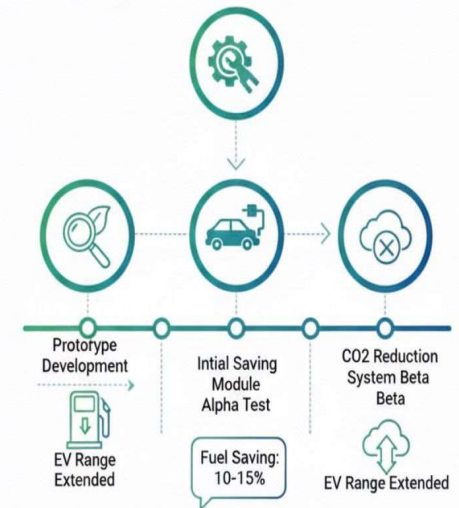
- **Social:** Safer & greener transport
- **Economic:** Saves fuel/electricity cost
- **Environmental:** Reduced emissions, sustainable driving
- **Industry Impact:** Scalable for cars, buses, EVs

## IMPACT & BENEFITS



Powering the Future of Mobility, Sustainably.

## Sustainable Transport Project Timeline



Advancing India's Green Future

- **IEEE/SAE** papers on energy harvesting suspensions
- **WHO/India** stats on fuel & energy usage in transport
- Existing patents & academic studies

